

Exercise #1: Benchmarking Throughput

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1. Measurement Methodology

To ensure accurate and consistent throughput calculations, the benchmarking logic utilizes two primary mechanisms:

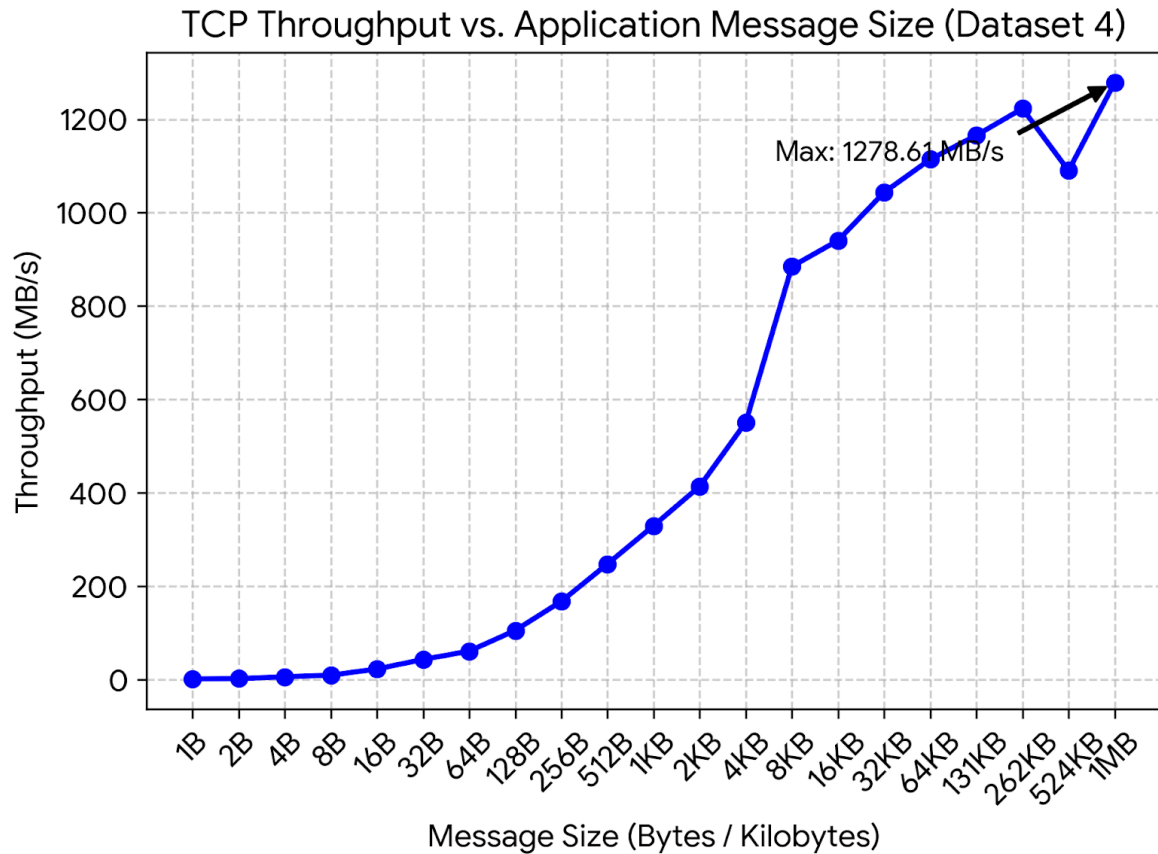
- **Warm-up Cycles:** Before starting the high-resolution timer for any specific message size, the client performs **100** warm-up cycles. This primes the network path, stabilizes the TCP window, and prevents initial network wake-up latency from skewing the timed results.
- **Dynamic Batch Sizing:** Rather than sending a fixed number of messages for every test, the client dynamically calculates the batch size so that exactly **256MB** of total data is transmitted per test case. Small packages (e.g., 1 byte) suffer heavily from protocol overhead and network noise, so sending a massive quantity of them averages out the noise and provides a stable measurement. Conversely, larger packages (e.g., 1MB) have less proportional overhead and take longer to transmit; sending a smaller, limited amount of them prevents the benchmark from taking an impractical amount of time while still yielding highly accurate results.

2. Performance Results

Raw Output Data:

```
Windows PowerShell
Unpacking objects: 100% (3/3), 271 bytes | 24.00 KiB/s, done.
From https://github.com/yitzchakkup/WorkshopCommunicationNetworks_EX1
  8f1c060..ec2b83d  main    -> origin/main
Updating 8f1c060..ec2b83d
Fast-forward
 client.c | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
yit.kup@mlx-stud-04:~/WorkshopCommunicationNetworks_EX1 $ ./client 10.128.128.213 9000
1      1.64    MB/s
2      2.45    MB/s
4      6.25    MB/s
8      9.54    MB/s
16     23.03   MB/s
32     43.35   MB/s
64     61.14   MB/s
128    105.58  MB/s
256    167.92  MB/s
512    247.88  MB/s
1024   329.80  MB/s
2048   414.10  MB/s
4096   551.25  MB/s
8192   884.77  MB/s
16384  940.53  MB/s
32768  1043.63  MB/s
65536  1114.26  MB/s
131072 1165.64  MB/s
262144 1223.01  MB/s
524288 1089.97  MB/s
1048576 1278.61  MB/s
yit.kup@mlx-stud-04:~/WorkshopCommunicationNetworks_EX1 $
```

Throughput Visualization:



3. Analysis

As illustrated in the graph, the measured TCP throughput follows a classic network saturation curve. At very small message sizes (1B to 64B), throughput is severely constrained; the fixed overhead of TCP/IP headers vastly outweighs the application payload, and the CPU is bottlenecked by the high volume of system calls required per byte transferred.

As the message size increases, the data-to-overhead ratio improves dramatically, resulting in a steep climb in transmission efficiency between 2KB and 64KB. As the payload approaches larger sizes, the network link approaches full saturation. The throughput subsequently reaches its absolute peak of 1278.61 MB/s at the 1MB message size, indicating the maximum saturated capacity of the network link (over our test cases).